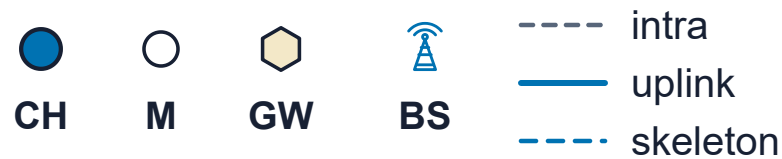
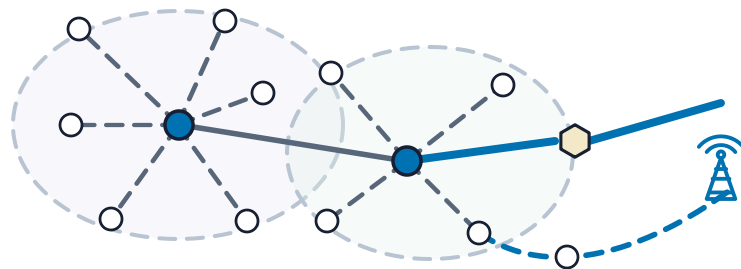


AERIS round



Phase 1 Cluster Formation



1

Round state



time slot



residual energy



neighborhood

2

CH scoring

Elect CH



energy



centrality



link quality

3

Cluster association



join CH



balance



update



Phase 2 Intra-Cluster Communication

Inputs: energy · link quality · density · buffer

1

CAS mode selector

select one mode

Direct

$M \rightarrow CH$



Chain

$M \rightarrow M \rightarrow CH$



Two-hop

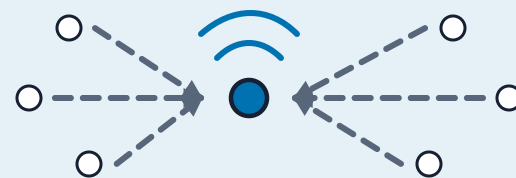
$M \rightarrow R \rightarrow CH$



○ M ⊕ R ● CH

2

Intra-cluster forwarding



3

CH aggregation



deduplicate



fuse



buffer



Phase 3 Uplink to BS

pre-transmission route selection

Q1

Direct acceptable? Yes

$LQ(CH, BS) \geq T_d$

Direct uplink



$CH \rightarrow BS$

No

Q2

Gateway valid? Yes

$Score_{GW} \geq T_g$

Gateway-assisted uplink



$CH \rightarrow GW \rightarrow BS$

main gain

No

Q3

Skeleton path valid? Yes

$Path_{skel} \neq \emptyset$

Skeleton reserve



$CH \rightarrow CH \rightarrow BS$

reserve

No

One-shot fallback



$CH \rightarrow BS$

single attempt